Experiment – 1

Solve 8 Puzzle program

Aim:

To develop a Python program to solve the 8-Puzzle problem using the Breadth-First Search (BFS) algorithm and find the shortest sequence of moves from the initial state to the goal state.

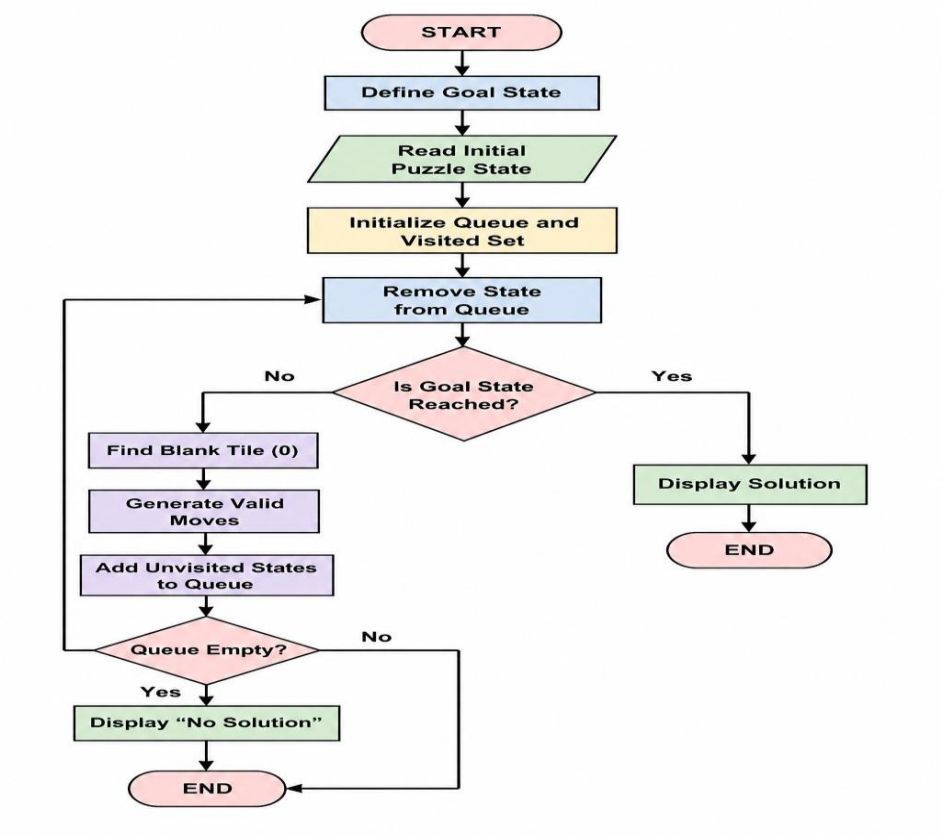
Objective:

- To represent the 8-puzzle problem using a 3×3 matrix.
- To implement the **Breadth-First Search (BFS)** algorithm in Python.
- To find the shortest sequence of moves from the initial state to the goal state.
- To demonstrate state-space search and problem-solving techniques in Artificial Intelligence.

Algorithm:

1. **Start** the program.
2. Define the **goal state** of the 8-puzzle.
3. Read the **initial state** from the user.
4. Create an empty **queue** and insert the initial state into it.
5. Create an empty **visited set** to store explored states.
6. Repeat until the queue is empty
7. If the queue becomes empty and the goal state is not reached, display **"No solution found."**
8. **End** the program.

Flowchart:



Code:

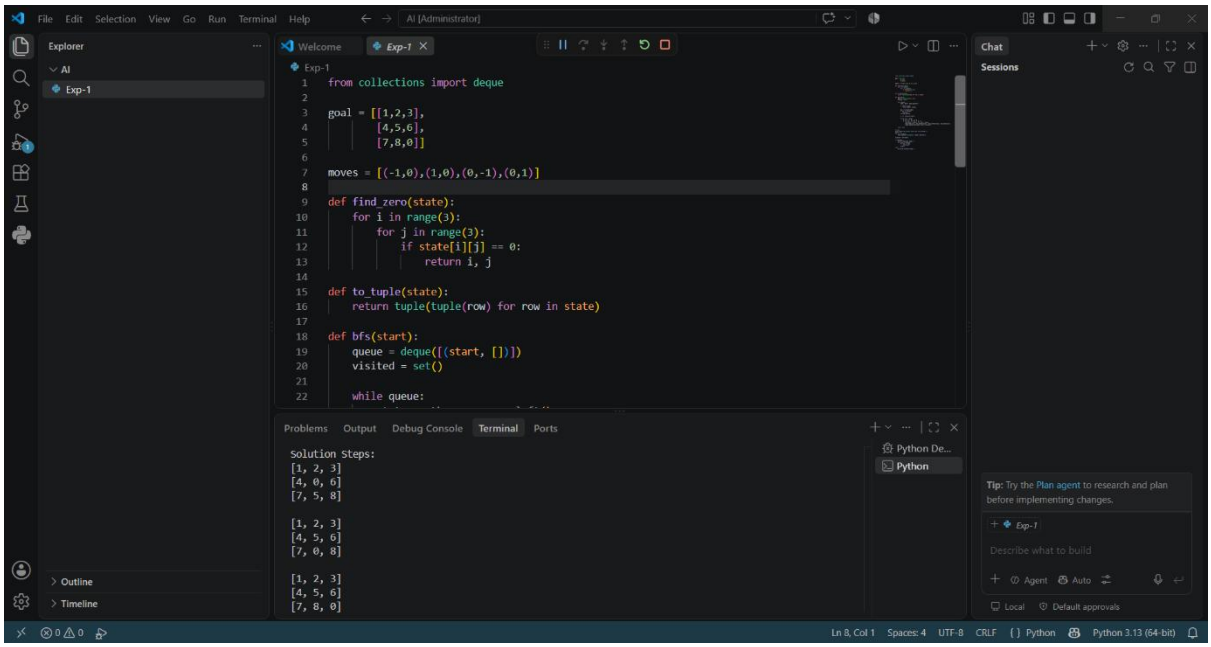
```
from collections import deque

goal = [[1,2,3],[4,5,6],[7,8,0]]
moves = [(-1,0),(1,0),(0,-1),(0,1)]

def bfs(start):
    q = deque([(start, [])])
    visited = set()

    while q:
        state, path = q.popleft()
        if state == goal:
```

```
            return path + [state]

        t = tuple(map(tuple, state))
        if t in visited:
            continue
        visited.add(t)

        for i in range(3):
            for j in range(3):
                if state[i][j] == 0:
                    x, y = i, j

        for dx, dy in moves:
            nx, ny = x+dx, y+dy
            if 0 <= nx < 3 and 0 <= ny < 3:
                s = [r[:] for r in state]
                s[x][y], s[nx][ny] = s[nx][ny], s[x][y]
                q.append((s, path+[state]))
    return None

print("Enter 3x3 puzzle:")
start = [list(map(int, input().split())) for _ in range(3)]

ans = bfs(start)

if ans:
    for s in ans:
        for r in s:
            print(r)
        print()
```

else:

print("No solution")

Output:



Result:

The 8-Puzzle problem was successfully solved using the Breadth-First Search (BFS) algorithm, and the shortest path from the initial state to the goal state was obtained.

Conclusion:

The 8-Puzzle problem was successfully solved using the Breadth-First Search (BFS) algorithm, demonstrating its effectiveness in finding the shortest path to the goal state.